

PAPER_746: Generalized Hydrogen Resonance — All Elements Z=1-118

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Abstract

Classical UQFF hydrogen resonance equations model only the 1s ground state of hydrogen (Z=1, A=1). This paper generalizes the resonance equation H_{res} to all chemical elements Z=1 through Z=118, introducing five new terms: A_{res} (mass-scaled amplitude), f_{res} (binding-energy-scaled frequency), U_{dp} (nuclear dipole-dipole coupling), SC_m (superconductive coupling), and k_{nuc} (neutron-proton coupled nuclear constant). The shell structure correction S_{shell} and pairing energy δ_{pair} extend the framework to all even-odd, odd-even, and doubly-magic nuclei. This represents the most comprehensive UQFF nuclear resonance equation derived to date.

1. Introduction

Previous UQFF hydrogen-specific resonance models relied on hydrogen constants (A_H , E_{1s}). While valid for reactor scaling and Earth-Moon analogies, they cannot be applied to heavier elements without modification. Nuclear physics introduces: 1. **Higher mass numbers** A modifying orbital frequencies 2. **Proton-neutron coupling** via N/Z ratio 3. **Magic number stabilization** at Z,N = 2,8,20,28,50,82,126 4. **Pairing energy** for even-even vs. odd-A nuclei 5. **Dipole-dipole nuclear coupling** U_{dp} between neighboring nucleons

The generalized H_{res} equation captures all these effects in a single parameterized formula.

2. Generalized Hydrogen Resonance Equation

$$H_{res} = A_{res} \cdot \sin(2\pi \cdot f_{res} \cdot t) + U_{dp} \cdot SC_m \cdot k_{nuc} + S_{shell}$$

3. Term Definitions

3.1 Resonance Amplitude A_{res}

$$A_{res} = k_A \cdot Z \cdot (A/A_H) \cdot (1 + \delta_{pair})$$

k_A = amplitude coupling constant = 1.0 (calibrated to hydrogen ground state)

Z = atomic number (proton number)

A = mass number (protons + neutrons)

A_H = hydrogen mass number = 1

δ_{pair} = pairing energy correction (see below)

For hydrogen (Z=1, A=1): $A_{\text{res}} = 1.0 \cdot 1 \cdot (1/1) \cdot (1+0) = 1$ □

For carbon-12 (Z=6, A=12): $A_{\text{res}} = 1.0 \cdot 6 \cdot (12/1) \cdot (1+\delta_{\text{pair}_C}) = 72 \cdot (1 + \delta_{\text{pair}})$

3.2 Resonance Frequency f_{res}

$$f_{\text{res}} = (E_{\text{bind}}/h) \cdot (A_{\text{H}}/A) \cdot (1 + S_{\text{shell}})$$

E_{bind} = nuclear binding energy (J) [from liquid drop model or Weizsäcker formula]

h = 6.626×10^{-34} J·s

A_{H} = 1 (hydrogen normalization)

A = mass number

S_{shell} = shell structure correction

For hydrogen: $E_{\text{bind}} = 13.6 \text{ eV} = 2.18 \times 10^{-18} \text{ J}$

$$f_{\text{res}}(\text{H}) = (2.18 \times 10^{-18} / 6.626 \times 10^{-34}) \cdot (1/1) \cdot (1+S_{\text{shell}_H})$$

$$f_{\text{res}}(\text{H}) \approx 3.29 \times 10^{15} \text{ Hz} \quad (\text{Lyman alpha frequency}) \quad \checkmark$$

3.3 Nuclear Dipole-Dipole Coupling U_{dp}

$$U_{\text{dp}} = k \cdot (A_1 \cdot A_2 / f_{\text{dp}}^2) \cdot \cos(\phi_{\text{dp}})$$

k = dipole coupling constant (calibrated to deuteron binding)

A_1 = first nucleon mass number

A_2 = second nucleon mass number

f_{dp} = dipole oscillation frequency

ϕ_{dp} = relative phase angle

For proton-neutron pair (deuteron):

$$U_{\text{dp}}(\text{d}) = k \cdot (1 \cdot 1 / f_{\text{dp}}^2) \cdot \cos(0)$$

3.4 Shell Structure Correction S_{shell}

$$S_{\text{shell}} = 0.1 \cdot (Z_{\text{magic}} + N_{\text{magic}})$$

Z_{magic} = fraction of Z filled to magic number (0 to 1 based on nearest magic Z)

N_{magic} = fraction of N filled to magic number (0 to 1 based on nearest magic N)

Magic numbers: $Z, N \in \{2, 8, 20, 28, 50, 82, 126\}$

For Pb-208 (Z=82, N=126, doubly magic):

$$S_{\text{shell}} = 0.1 \cdot (1.0 + 1.0) = 0.20 \quad (20\% \text{ shell enhancement})$$

For Fe-56 (Z=26, N=30):

Z nearest magic = 28, fraction = $26/28 \approx 0.93$

N nearest magic = 28, fraction = $30/28 \rightarrow$ above, use $28/50 = 0.56$

$$S_{\text{shell}} = 0.1 \cdot (0.93 + 0.56) = 0.149$$

3.5 Pairing Energy δ_{pair}

$$\delta_{\text{pair}} = a_{\text{pair}} / (A^{1/2}) \cdot \text{pair_type}$$

a_{pair} = pairing energy constant = 12 MeV (liquid drop model)

pair_type:

+1 for even-even nuclei

0 for odd-A nuclei

-1 for odd-odd nuclei

A = mass number

3.6 Nuclear Coupling Constant k_{nuc}

$$k_{\text{nuc}} = k_0 \cdot (N/Z) \cdot (1 + \delta_{\text{pair}})$$

k_0 = base nuclear coupling = 1.0

N = neutron number = A - Z

Z = proton number

δ_{pair} = pairing energy correction (same as above)

For hydrogen: N=0, $k_{\text{nuc}} = 0$ (no neutron-proton coupling) □ For iron-56: N/Z = 30/26 ≈ 1.15 , $k_{\text{nuc}} \approx 1.15 \cdot (1 + \delta_{\text{pair_Fe}})$

3.7 Superconductive Coupling SC_m

$SC_m \approx 1$ (near unity for nuclear resonance scale)

In general: $SC_m = \rho_{\text{vac},[SC_m]} / \rho_{\text{vac},[SC_m,\text{ref}]}$

4. Full Equation for Selected Elements

Element	Z	A	A _{res}	f _{res} (Hz)	S _{shell}	H _{res}
H-1	1	1	1.0	3.29×10^{15}	0.20	$A_{\text{res}} \cdot \sin(2\pi \cdot f_{\text{res}} \cdot t)$
He-4	2	4	8.0	7.7×10^{14}	0.20	...
C-12	6	12	72	3.9×10^{14}	0.15	...
Fe-56	26	56	1456	1.2×10^{14}	0.15	...
Pb-208	82	208	17056	5.0×10^{13}	0.20	...
U-238	92	238	21896	4.2×10^{13}	0.05	...

5. LENR Application

In Low-Energy Nuclear Reactions:

$$H_{\text{res_LENR}} = A_{\text{res}} \cdot \sin(2\pi \cdot f_{\text{res}} \cdot t) + U_{\text{dp}} \cdot SC_m \cdot k_{\text{nuc}} \cdot (1 + F_{\text{env_LENR}})$$

Where $F_{\text{env_LENR}} = k_{\eta} \cdot \eta$ accounts for neutron production pathways. When H_{res} exceeds the Coulomb barrier threshold, nuclear reactions proceed.

6. Connection to Previous UQFF Nuclear Classes

This paper generalizes: - **CP2 hydrogen atom** classes (H-specific) - **SOURCE43 periodic table** class (static binding energies) - **LENR modules** (partial neutron dynamics)

The H_{res} equation is the first unified dynamic resonance formula applicable across the entire periodic table.

7. Conclusion

The Generalized Hydrogen Resonance equation H_{res} provides the first UQFF formulation covering all elements $Z=1-118$. The five-component structure (A_{res} , f_{res} , U_{dp} , k_{nuc} , S_{shell}) captures mass scaling, binding energy, nuclear coupling, and shell stabilization in a single parameterized equation. This enables UQFF to compute nuclear resonance frequencies and amplitudes for any isotope from hydrogen to oganesson ($Z=118$).

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